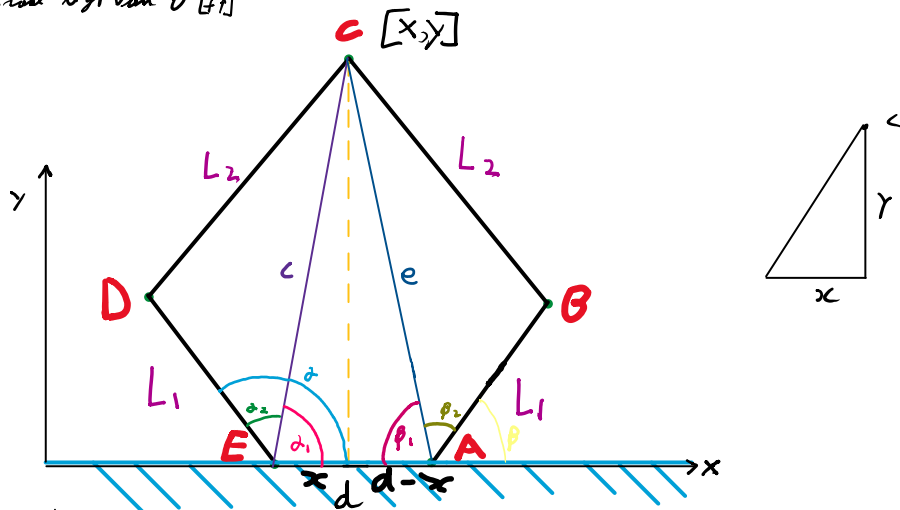


Positie van eindpunt C wordt beschreven als $C[x, y]$
 x - Hoe ver C horizontaal ligt van E
 y - Hoe ver C verticaal ligt van D [ft]



$$c = \sqrt{x^2 + y^2}$$

$$e = \sqrt{(d-x)^2 + y^2}$$

$$\alpha_1 = \tan^{-1}\left(\frac{y}{d-x}\right) = \tan^{-1}\left(\frac{y}{x}\right)$$

(Cosinus Regel)

$$\begin{aligned} \alpha_2 &= \cos^{-1}\left(\frac{L_1^2 + c^2 - L_2^2}{2 \cdot L_1 \cdot c}\right) \\ &\Rightarrow L_2^2 = L_1^2 + c^2 - 2 \cdot L_1 \cdot c \cdot \cos(\alpha_2) \\ &\Rightarrow L_2^2 - L_1^2 - c^2 = -2 \cdot L_1 \cdot c \cdot \cos(\alpha_2) \\ &\Rightarrow \frac{L_2^2 - L_1^2 - c^2}{-2 \cdot L_1 \cdot c} = \cos(\alpha_2) \\ &\Rightarrow \alpha_2 = \cos^{-1}\left(\frac{L_2^2 - L_1^2 - c^2}{-2 \cdot L_1 \cdot c}\right) \end{aligned}$$

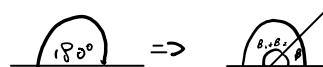
$$\alpha = \alpha_1 + \alpha_2$$

$$\beta_1 = \tan^{-1}\left(\frac{y}{d-x}\right) = \tan^{-1}\left(\frac{y}{d-x}\right)$$

(Cosinus Regel)

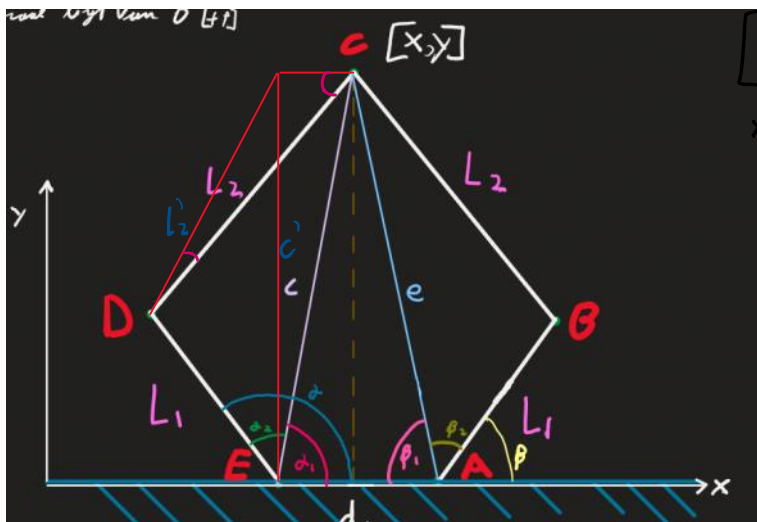
$$\begin{aligned} \beta_2 &= \cos^{-1}\left(\frac{L_1^2 + e^2 - L_2^2}{2 \cdot L_1 \cdot e}\right) \\ &\Rightarrow L_2^2 = L_1^2 + e^2 - 2 \cdot L_1 \cdot e \cdot \cos(\beta_2) \\ &\Rightarrow L_2^2 - L_1^2 - e^2 = -2 \cdot L_1 \cdot e \cdot \cos(\beta_2) \\ &\Rightarrow \frac{L_2^2 - L_1^2 - e^2}{-2 \cdot L_1 \cdot e} = \cos(\beta_2) \\ &\Rightarrow \beta_2 = \cos^{-1}\left(\frac{L_2^2 - L_1^2 - e^2}{-2 \cdot L_1 \cdot e}\right) \end{aligned}$$

$$\beta = 180^\circ - \beta_1 + \beta_2$$



- Gegeven (input) → gewenst punt van de end effector C dus coördinaten x, y

- Gezocht (output) → de gewrichtshoeken ($\alpha_1, \alpha_2, \beta_1, \beta_2$) die naar de servo gestuurd worden

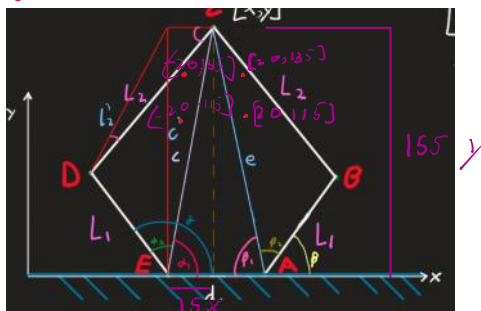


$$[X, Y] \rightarrow [\alpha, \beta]$$

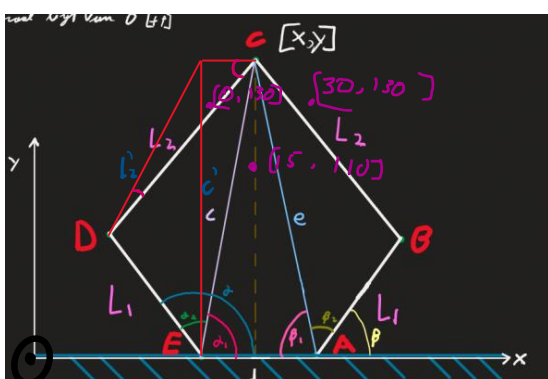
$$x, y \rightarrow c, e, \alpha_1, \beta_1$$

Voor jouw 5-bar in 2D doen we:

- Input: gewenst punt $C(x, y)$.
- Output: twee servohoeken (bijv. alpha links, beta rechts).



$$\begin{aligned} & (15, 150) \\ & \downarrow \\ & [-20, 135] \rightarrow [20, 135] \\ & \downarrow \\ & [-20, 115] \rightarrow [20, 115] \end{aligned}$$



2

Körper Vorrat $\partial(-a)$

